

JEE MAIN REVISION MASTER: CIRCULAR MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 7 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kinematics of Circular Motion: Angular Velocity & Acceleration	4 PYQs	Kinematics
Moderate	Dynamics: Centripetal Force, Banking & Turntable Slipping	3 PYQs	Dynamics & Problem Archetypes

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Kinematics of Circular Motion: Angular Velocity & Acceleration	$v = r\omega$, $a_c = \frac{v^2}{r} = \omega^2 r = 4\pi^2 f^2 r$, $a_{\text{net}} = \sqrt{a_c^2 + a_t^2}$
Dynamics: Centripetal Force, Banking & Turntable Slipping	$F_c = \frac{mv^2}{r} = m\omega^2 r$, $r_1 \omega_1^2 = r_2 \omega_2^2$, $\tan\theta = \frac{v^2}{rg}$, $v_{\text{max}} = \sqrt{\mu rg}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kinematics of Circular Motion: Angular Velocity & Acceleration (Moderate Yield — 4 PYQs)

Core Principle: Circular motion kinematics: $v = \omega r$, centripetal acceleration $a_c = (v^2 / r) = \omega^2 r$. In uniform circular motion (UCM), speed and kinetic energy are constant, but velocity and acceleration vectors continuously change direction.

■ **Caution / JEE Pitfall:** After one complete revolution in UCM, total distance is $2\pi r$, but net displacement is zero!

■ **Speed Trick:** In UCM, work done by the centripetal force is always identically zero ($W = 0$) because $\vec{F}_c \perp \vec{v}$.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The ..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "An athlete runs along a circular track of diameter 80 m. The distance travelled and the magnitude of displacement of the athlete when he co..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?..." [Ans: N/A]

3.2 Dynamics: Centripetal Force, Banking & Turntable Slipping (Moderate Yield — 3 PYQs)

Core Principle: Centripetal force $F_c = (m v^2 / r) = m \omega^2 r$. Slipping on a rotating turntable: $\mu m g = m \omega^2 r$ implies $\omega^2 r = \text{const}$. Angle of banking for frictionless turning: $\tan \theta = (v^2 / r g)$.

■ **Caution / JEE Pitfall:** For turntable coin problems, distance r is inversely proportional to ω^2 ($r \propto 1/\omega^2$). If ω is doubled, coin must be placed at $1/4$ the original distance to avoid slipping.

■ **Speed Trick:** For a suspended pendulum bob inside a car rounding a curve of radius r at speed v , the string deflects by angle $\tan \theta = (v^2 / r g)$ from the vertical.

Representative Source PYQs:

- **[2022 Shift] Q#4 (MCQ):** "A car is moving in a circular horizontal track of radius 10m with a constant speed of 10ms^{-1} . A bob is suspended from the roof of the car by a light wire of length 1.0m. The angle made by the wire with the vertical is (in rad)" **[Ans: N/A]**
- **[2021 Shift] Q#1 (MCQ):** "A coin placed on a rotating turn table just slips if it is placed at a distance of 4cm from the centre. If the angular velocity of the turntable is doubled, the coin must be placed at what distance from the centre to avoid slipping?" **[Ans: N/A]**
- **[2020 Shift] Q#2 (MCQ):** "One end of a string of length l is connected to a particle of mass m and the other to a small peg on a smooth horizontal table. If the particle is moving in a circular path of radius r , the angle made by the string with the vertical is" **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The speed of the particle is 2ms^{-1} , when the particle attains the lowest 'y' co-ordinate, the acceleration of the particle is (in ms^{-2})

- (A) 0.4j
- (B) 0.8i
- (C) 0.8j
- (D) 0.4i

Correct Answer: None

Solution Analysis: To determine the acceleration of a particle in uniform circular motion, we start with the equation of the circle: $(x-2)^2 + y^2 = 25$. This equation represents a circle with its center at $(2,0)$ and a radius r of 5 meters, derived from: $r^2 = 25 \Rightarrow r = 5$. In uniform circular motion, the centripetal acceleration a_c is given by the formula: $a_c = \frac{v^2}{r}$. Substituting the given speed $v = 2\text{m/s}$ and the radius $r = 5\text{m}$: $a_c = \frac{2^2}{5} = 0.8\text{m/s}^2$. When the particle is at the lowest y-coordinate, it is at the point $(2, -5)$, assuming

Benchmark Problem 2: KCET 2022 (Q#4)

A car is moving in a circular horizontal track of radius 10m with a constant speed of 10ms^{-1} . A bob is suspended from the roof of the car by a light wire of length 1.0m. The angle made by the wire with the vertical is (in rad)

- (A) $(\pi / 4)$
- (B) zero
- (C) $(\pi / 3)$
- (D) $(\pi / 6)$

Correct Answer: None

Solution Analysis: Radius of circular track, $r = 10\text{m}$ speed of car, $v = 10\text{m/s}$. The given situation is shown below. $T \sin \theta = \frac{mv^2}{r}$ (i) $T \cos \theta = mg$ (ii) Dividing Eq. (i) by Eq. (ii), we get $(T \sin \theta / T \cos \theta) = \frac{mv^2}{rmg} \Rightarrow \tan \theta = \frac{v^2}{rg} = \frac{10^2}{10 \times 10} = 1 \Rightarrow \tan \theta = \tan 45^\circ \Rightarrow \theta = 45^\circ = (\pi / 4)\text{rad}$